

Reviewer Pack — Minimal Monoidal Categorical Generators and Communication Co...

Entry 3a2810a232ec · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Monoidal Categorical Generators and Communication Complexity Rank

Entry ID: 3a2810a232ec **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-04 05:18:55 UTC

1. Conjecture

Field A (mathematical branch): Monoidal Category Theory **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For every boolean function f , the minimal number of generators in the monoidal category associated with its communication complexity is linearly correlated with its communication complexity rank, such that $|G(f)| = \Theta(\text{rank}(f))$, where $G(f)$ denotes the set of minimal monoidal categorical generators for the function f .

Rationale (proposer's reasoning):

Monoidal category theory provides a framework to study algebraic structures through diagrams, which could offer new insights into the structure of communication complexity. By examining the minimal generators in this category, we may uncover fundamental properties that are not apparent from traditional analysis, potentially revealing deep connections between categorical structures and computational complexity.

Taxonomy category: Monoidal_Categorical_Generators (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): d2f75b3877111b18

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between the size of the minimal monoidal categorical generators and the communication complexity rank, across 30 randomly seeded instances, is ≥ 0.8 AND the mean time to compute the generator set for $n \leq 40$ does not exceed 240 seconds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity_rank(f):
        n = len(f)
        max_rank = 0
        for i in range(1 << (n - 1)):
            rank = 0
            for j in range(n):
                if f[i ^ (1 << j)] != f[i]:
                    rank += 1
            max_rank = max(max_rank, rank)
        return max_rank

    def minimal_monoidal_categorical_generators(f):
        n = len(f)
        generators = set()
        for i in range(1 << n):
            if all(f[i ^ (1 << j)] != f[i] for j in range(n)):
                generators.add(i)
        return generators

    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        f = generate_boolean_function(n)
```

```

rank = communication_complexity_rank(f)
generators = minimal_monoidal_categorical_generators(f)

if len(generators) > 100: # Avoid excessive computation
    continue

results.append({
    "metric_name": "communication_complexity_rank",
    "metric_value": rank,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": False,
    "counterexample": ""
})

if not results:
    return {
        "metric_name": "communication_complexity_rank",
        "metric_value": None,
        "instances_tested": 0,
        "n_max": 0,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

mean_metric = sum(r["metric_value"] for r in results) / len(results)
correlation_coefficient = None

return {
    "metric_name": "communication_complexity_rank",
    "metric_value": mean_metric,
    "instances_tested": len(results),
    "n_max": max(r["n_max"] for r in results),
    "conjecture_holds": False,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if not r["conjecture_holds"]) / len(results)

    if all(not r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])

```

```
print(f"RESULT: FALSIFIED counterexample='' first_failing_seed={first_failing_seed}")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a3dc230d.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a3dc230d.py", line 48, in run_trial
    rank = communication_complexity_rank(f)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a3dc230d.py", line 30, in communication
    if f[i ^ (1 << j)] != f[i]:
       ~^^^^^^^^^^^^
```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete the computation to determine the Pearson correlation coefficient and mean computation time. | next: Re-run the test with proper error handling to ensure it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13986
2	propose	ollama_remote	glm4:latest	0	0	12319
3	preregistration	ollama_remote	glm4:latest	0	0	37268
4	novelty	ollama_remote	glm4:latest	0	0	8546
5	novelty	ollama_remote	glm4:latest	0	0	11588
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16231
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9984
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14080
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10373
10	judge	ollama_remote	glm4:latest	0	0	11397

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 145773 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/3a2810a232ec.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/3a2810a232ec.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/3a2810a232ec.tar.gz (if generated)